

[TEGMARK] Cosmology blind-pin bar — PRE-REGISTERED before the domain opens

Grace | 2026-07-31 | the SM sector is complete; cosmology is the next domain (Casey's "reinvestigate cosmology after the SM" plan, triggered). Cal's flag: cosmology is the highest target-awareness-risk domain — and this session LIVED there (the DE drift, -0.949 fit, three times). So I commit the bar NOW, blind, before any cosmology number lands, so it cannot be retrofitted. This is the -0.949 lesson turned into a standing gate for the whole domain.

Why the bar binds hardest here (the domain's specific danger)

Cosmology is dense with cheap rationals (Ω fractions, w -values, N_e , z_{eq} , n_s , spectral indices) and flexible/anomaly datasets (CMB — "quote-anything"; DESI DE, moving). The fit-temptation is maximal and the data moves under you. Every failure mode we caught this session concentrates here: cheap-rational fits (θ_{23} 4/7), stale-experimental-number drift (Σm_ν , the DESI w_0), model-conditional bounds (Λ CDM caveat), and coincidence-welds ($c_2=11$, $n_f=6=C_2$).

★ THE BAR (committed blind, before the domain opens)

1. Blind-pin every rational before the datum. Any cosmology rational (Ω_Λ , w , ratios, exponents) must have its integer sourced from the geometry BEFORE the observed value is consulted. A rational that fits but isn't blind-pinned does NOT rule Derived — the cosmology window is dense with cheap rationals (the θ_{23} lesson, maximal here).
2. ★ cc-magnitude = FIT-SUSPECT until a mechanism FORCES it (Cal). Λ 's smallness ($\sim 10^{-122}$) is the deepest prize and the most fit-temptable. The banked form $\Lambda = g \cdot \exp(-C_2(g^2 - \text{rank})) = 7 \cdot \exp(-282)$ (pred_089) is a candidate: matching 122 orders by an exponent is NOT a derivation until a mechanism forces the exponent 282. The magnitude does NOT bank on a matching exponent — require the substrate-coupling mechanism (Λ 's smallness = the substrate coupling's smallness). Treat as fit-suspect; the exponent is the open piece.
3. DE is $w=-1$, banked (K1040) — do NOT re-open the -0.949 fit. The $\varepsilon(a)/w(a)$ forward derivation (task #54, from the Bergman matched-fraction) must FORCE any deviation from $w=-1$, not fit it to DESI. The dynamical forms (-0.949 T2079, -0.99973) are RETIRED (superseded). A dynamical $w(a)$ claim banks only if the geometry forces it target-innocently; "matches the DESI dynamical-DE hint" is the -0.949 trap, not evidence.
4. CMB is quote-anything — target, not evidence (my standing memory). CMB anomalies are $\sim 2-3\sigma$, cosmic-variance-consistent, fit a dozen rival stories. Admissible use = a specific, distinctive, PRE-REGISTERED signal only (e.g. a bubble-collision scar profile vs Feeney-Johnson-Peiris), committed before looking. Post-hoc "consistent with the Axis of Evil" $\approx$ worthless.
5. Score σ , not %; account for look-elsewhere. Every cosmology match scored against the experimental error, carried scheme-aware; look-elsewhere across the many cosmology observables. Verify the CURRENT fit (DESI DR2, Planck, ACT), not remembered numbers — the $\Sigma m_\nu/w_0$ drift lesson.
6. Provenance-not-value (Rule 11), applied to scales and states. Two cosmology numbers sharing a value are distinct if different operators/senses produce them (the $c_2=11$ and mass-gap lessons). Don't weld a coincidence into a mechanism.

The parked forward work this gates

- $\varepsilon(a) / w(a)$ derivation (task #54): forced deviation from $w=-1$, or none — bar #1, #3.
- cc-magnitude prize (the deep one): the exponent-282 mechanism — bar #2 (fit-suspect until forced).
- Cosmology-sector reconciliation: the two w_0 forms (retired), Λ/Ω ratios ($\Omega_\Lambda=13/19$ blind-pin check), z_{eq} , n_s — bar #1, #5.

Handoff

- Me: hold this bar as each cosmology number lands; blind-pin audit + σ -score + provenance check; catalog only when the bar clears.
- @Lyra/@Elie: $\varepsilon(a)$ and cc-magnitude are forward derivations — the geometry forces or it doesn't; a match to moving DE data is not a pass.
- @Cal/@Keeper: the bar is committed here, blind, before the domain — verify against it, not against whatever lands.

— Grace, 2026-07-31 [TEGMARK]. Cosmology blind-pin bar PRE-REGISTERED before the domain opens (Cal's highest-target-awareness flag; this session lived the -0.949 drift $3\times$): (1) blind-pin every rational before the datum; (2) cc-magnitude FIT-SUSPECT until the exponent-282 mechanism forces it (pred_089 is candidate); (3) DE= $w=-1$ banked, don't re-open -0.949 , force any $w(a)$ deviation not fit it; (4) CMB quote-anything = target not evidence, pre-registered scar only; (5) score σ not %, verify current fits; (6) provenance-not-value. Committed blind so it can't be retrofitted.

★ $\varepsilon(a)$ AUDIT CRITERIA — committed BLIND before Elie's derivation (task #54, the first cosmology piece)

Grounding (reconnect): DE is $w=-1$ from the fixed $C\cdot\pi^5$ volume (K1040); the deviation is a vanishing substrate coupling $\varepsilon(a)$. Task #54 = derive $w(a)$ from the Bergman matched-fraction, $\varepsilon(a)$ the substrate coupling vs scale a . Note K1034: the dynamical-DE 'breathing-mode improvement' already FAILED ($640\times$ too small) $\rightarrow$ dynamical-DE-as-a-win is retired; BST is committed to $w=-1$, Σm_ν a genuine edge-kill.

The $\varepsilon(a)$ attempt PASSES (target-innocent) iff ALL: 1. Provenance: the matched-fraction / $\varepsilon(a)$ form is derived from the Bergman geometry, NOT selected to match DESI's w_0 (-0.84) or any DE datum. Blind — Elie derives $\varepsilon(a)$ BEFORE looking at DESI. 2. $w=-1$ consistency: $\varepsilon(a) \rightarrow 0$ (substrate coupling vanishes) $\rightarrow w \rightarrow -1$ (K1040). Any deviation from $w=-1$ must be FORCED by the matched-fraction, not fit. 3. ★ Relapse red-flag: if $\varepsilon(a)$ reproduces the RETIRED -0.949 (T2079) or -0.99973 by construction, that is the -0.949 relapse — a RED FLAG, not a win (those forms were target-aware, retired K1040/K1034). Landing on the DESI dynamical hint is the trap, not evidence. 4. σ + current data: score $w(a)$ against DESI DR2's error (not %); verify the current fit, not remembered numbers. 5. Keep it separate from the cc-magnitude: $\varepsilon(a)$ is the equation-of-state SHAPE; the cc-MAGNITUDE (Λ smallness, exponent-282) is a distinct fit-suspect piece — don't let one launder the other. VERDICT rule: forced target-innocent ($\varepsilon(a)$ from the geometry, $w \rightarrow -1$, deviation forced) $\rightarrow$ the deviation banks at its honest tier; fit to DESI / lands on the retired $-0.949 \rightarrow$ NOT a pass (candidate/relapse). I rule it the moment it lands.